

Can a breeder improve two traits at once?

Learning intention

Explain selective breeding and justify a breeding plan using inherited traits, performance evidence and genetic diversity.

Mission: plan a productive, disease-resistant crop.

A good choice must use evidence and survive a second check.

Retrieve inheritance

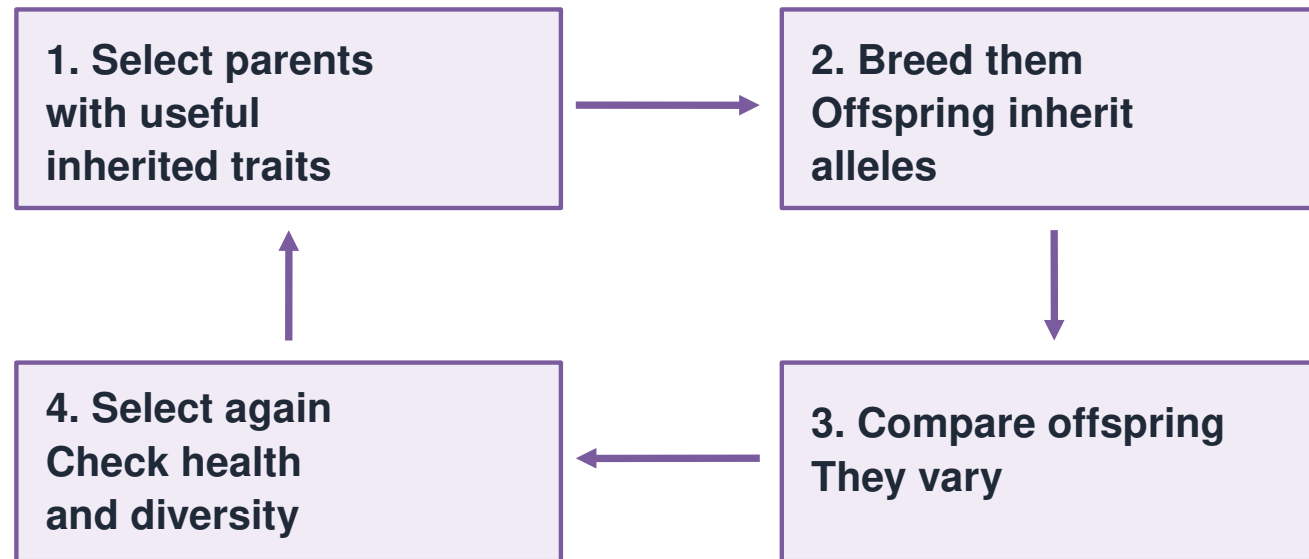
R1. What is an allele?

R2. Are sexual offspring always identical to their parents?

R3. Can growing conditions affect measured yield?

How selective breeding works

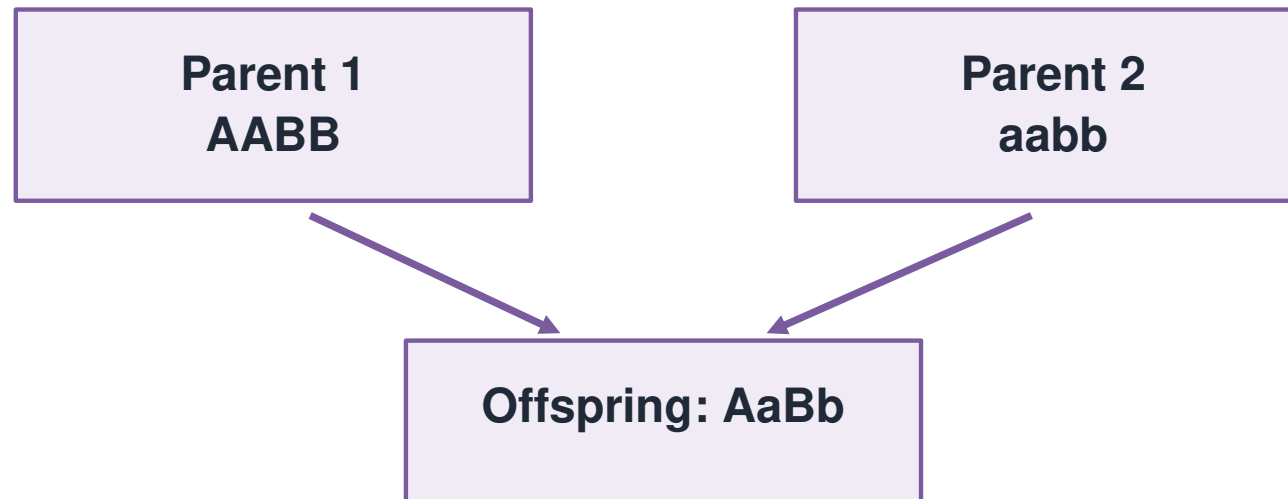
Select, breed, compare, repeat



Humans decide which organisms reproduce. Choose inherited traits, breed, compare offspring, select again. Repeat across generations.

Inheritance is a mechanism

An allele can be inherited; a wish cannot



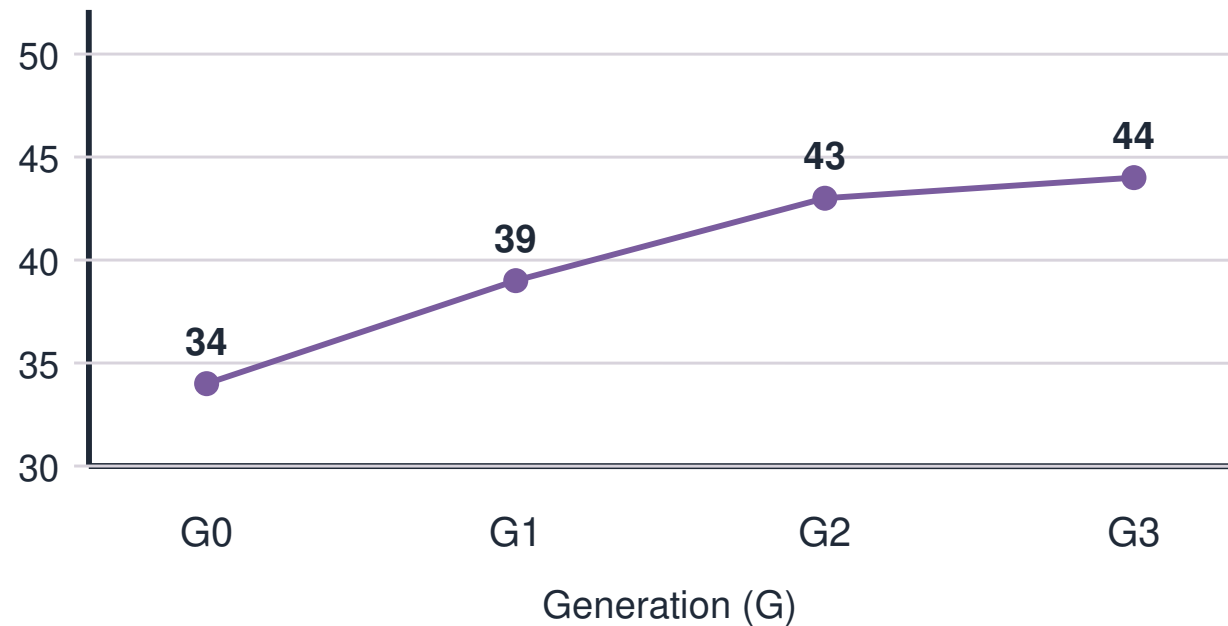
Symbol model only. Real yield and resistance can involve many genes and the environment.

Offspring receive alleles from both parents. A simple model can show inheritance. Real yield and resistance can involve many genes.

Read change across generations

Constructed sample data: selected crop line

Mean grain mass per plant (g); matched conditions



The population mean can shift after selection. G0 to G3: 34 g to 44 g per plant. A trend is not a guarantee for every plant.

Protect useful variation

Select useful traits and protect diversity



Neither plan guarantees health or future yield.
Family labels are a simplified model.

Repeated close-relative breeding can reduce diversity. Some harmful recessive combinations become more likely. Retain suitable unrelated lines and check offspring.

Breeding round 1: choose the parents

W1. Both parents: damage at most 2. At least one parent: yield at least 45 g. Use different family lines. Choose and justify.

Line / family	Mean grain mass (g/plant)	Disease damage (0–10)
A / F1	48	6
B / F2	40	2
C / F3	46	1
D / F1	52	8

Constructed sample data: matched trial conditions; lower damage is better. Each line mean comes from 10 plants.

Breeding round 2: respond to new evidence

W2. Offspring target: yield at least 45 g; damage at most 2. W3. Choose its next partner: sibling P or unrelated E? Justify the choice and one needed check.

Candidate / family	Grain mass (g/plant)	Disease damage (0–10)
P / F4	45	1
Q / F4	48	6
R / F4	42	0
S / F4	50	2
E / F5 (unrelated)	45	1

Constructed sample data: P–S are siblings from $B \times C$; E is from a separate suitable line. One trial per plant.

What changes after selection?

A breeder repeatedly selects resistant plants. Choose the best biological explanation.

- A** Each plant decides to change its genes.
- B** Desired inherited traits can become more common in later generations.
- C** Every offspring has exactly the same traits.
- D** Growing conditions no longer matter.

Write a two-generation breeding log

Use the shared candidate and offspring evidence.

Record selection, rejection and the reason for each.

Finish with a prediction and a test.

A decision needs its evidence

I selected ... because the table shows ...

I rejected ... because ...

I would check ... before the next generation.

Do not promise a perfect crop

Predict a possible shift in the population.

Keep uncertainty about individual offspring.

Name a way to retain diversity.

Peer-check the breeding plan

Can your reader follow both selections?

Ask one question about a trait or risk.

Use the response to improve one decision.

A strong breeding log

Names a valid selection with two data points.

Explains inheritance, variation and repeated selection.

Includes diversity and a practical next check.

Transfer the method

- E1. Give the four main selective-breeding steps.
- E2. Why test offspring before choosing again?
- E3. Name one risk of repeatedly using close relatives.